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MEM 426, METRIC ASSEMBLY, INSTALLATION AND COMMISSIONING

PROGRAMME: HND II POWER PLANT MAINTENANCE

SUMMARY NOTES.

MATERIAL HANDLING: This is an art and science involving the Moving, Packing and Storing of Substance in any form. It is the preparation, placing and positioning the material to facilitate their Movement or Storage.

OBJECTIVES OF MATERIAL HANDLING.

1. Reduction in wastage of Machine time and regular Supply of raw materials.
2. Reduction in Manufacturing cycle time
3. Avoiding disruption in production Schedule
4. Ensuring safety to workers and provide for safe working conditions.

GENERAL PRINCIPLES OF MATERIAL HANDLING.

- (i) Materials should be moved over shortest distance possible. Short Movements require less time and cost less money than long Movements
- (ii) Don't retrace your path. The aim of Material handling system should be to move in forward direction or clockwise direction. It is ~~impossible~~ time should be kept as short as possible. It is time taken for picking up the finished product.
- (iv) load should be carried both ways on Material Handling trip whenever possible.
- (v) Partial loads and Manual handling should be avoided
- (vi) utilize gravity as Moving force as far as possible.
- (vii) utilize containers and chute loads. The chute loads means the product to be moved, should be grouped into units of a large and constant size.
- (viii) Material handling should be marked already to help save time in Sorting and avoiding Moving up.

STONE KEEPING.

The purpose of Stone Keeping is to see that the stock of no item at anytime falls below the prescribed minimum or goes in excess of fixed maximum.

FUNCTIONS OF MAINTENANCE STORES.

- (i) Receiving, Storing and issuing and booking the goods or spare parts. Returning faulty goods when necessary.
- (ii) Storing stock in conditions which protects the spare parts stock from damage, deterioration and pilfering.
- (iii) Issue stock and keeping records of stock availability and movement using stock level control consumables, Spares and Materials.

STORE REQUISITION FORMAT / TABLE.

S/NO	MATERIAL	QTY	SUPPLIED	STORE	RATE	SIGN
1	Engine oil	20 ltrs	20 ltrs	Yes	15,000.00	M.C. - James
2	Gear oil	25 ltrs	25 ltrs	Yes	40,000.00	✓
3	Grease	0.8 kg	0.8 kg	Yes	10,000.00	✓
4	Cutting tool	2 pieces	2 pieces	Yes	11,600.00	✓
5	Boring tool	2 pieces	2 pieces	Yes	17,000.00	✓
6	Thread cutting tool	2 pieces	2 pieces	Yes	24,000.00	✓
7	Drum	20 ltrs	20 ltrs	Yes	37,000.00	✓
8	Rags	0.5 kg	0.9 kg	Yes	5,000.00	✓
TOTAL					157,000.00	

The personnel will fill the requisition as can be seen above and send it to the Store Keeper. The Store Keeper will issue those items on the card and reflect it on the store card.

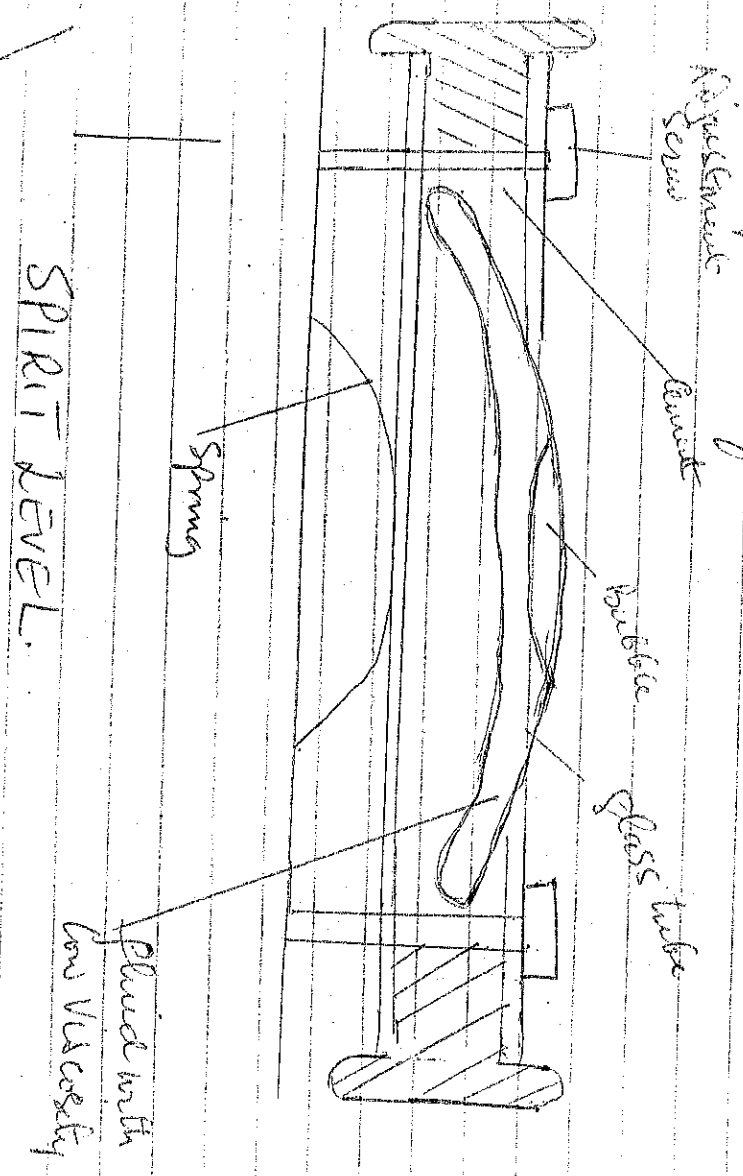
SPIRIT LEVEL.

A Spirit level is a glass tube, sealed at both ends, the inside of which is ground to a circular curve horizontally and vertically and nearly filled with spirit or liquid.

went to your body,

(iii) KEEP HANDS: Putting down the load is the exact reverse of your lifting procedure. At precise positioning of the load is required put it down, then slide it to the desired position.

Glass is alcohol allowing enough space to form a bubble. The liquid must be quite acting on rising and stable under normal temperature vibration. The radius of the circular longitudinal of the tube varies according to use. A very short radius makes a very sensitive bubble while a long radius makes a very sensitive bubble. It is important that the curve should be exactly circular so that equal distances on the tube subtend equal angles at the center of curvature. The spirit level is provided with a scale having uniform graduations etched on the exterior surface of the tube to show the exact position of the bubble.



THEODOLITE.

This is an instrument used to measure horizontal and vertical angles prolonging a straight line and vertical distances and to determine the direction of true north; it may be classified as Simple Vernier theodolite, Micro meter theodolite, optical theodolite and electronic theodolite. In level tubes are provided in the horizontal circle upper plate for leveling the theodolite quickly. One is placed parallel and the other at right angles to the line of sight.

FORMWORK.

Formwork is a mould which retains the shape, size and position of mixed concrete which is in form of fluid or wet concrete. It is possible with sand forms of concrete foundations to use the sides of the excavation as mould but in more cases when casting concrete members, a mould has to be constructed on site. These moulds are usually called formwork.

Formwork is necessary during installation because it offers all necessary resistance to the imposed pressures as a single member. The formwork sides retain the wet concrete in the required shape and able to withstand sides the initial hydrostatic pressure of the wet concrete. Formwork retaining the concrete and finally set the concrete until it has gained sufficient strength to be self supporting. Column formwork consist of a vertical mould of the desired shape and size which has to retain the wet concrete and resist the initial hydrostatic pressure caused by the wet concrete.

(2)

FACTORS CONSIDERED FOR A FOUNDATION TYPE

- (i) Soil condition
- (ii) Types of Machines to be installed
- (iii) Structural loading
- (iv) Economic factor
- (v) Time factor
- (vi) Weight of the Machine

GRIEVANCE.

Grievance is any human problem relating to the employment of an individual which remains him and which he thinks the authorities cannot solve. Grievance could be dissatisfaction experienced by workers through complaints and protests. Grievance could come from individual small groups working together or fully organised section of the workforce.

CAUSES OF GRIEVANCE.

(i) Alleged ill-treatment of a worker, generally a union member. This may be in the area disciplining measures, promotion, annual increments, distribution of overtime, wage payment, introduction to change, bonus, selection of group leader, security rights and query already given.

(ii) Personal problems to which workers are unable to adjust eg poor health, family illness, marital discord, financial difficulties etc. The frustrating result from these problems may cause the employees to find fault with his job, with others around him and the working environment.

(iii) Interpretation of union/management agreement - collective bargaining.

STAGES OF SOLVING INDIVIDUAL GRIEVANCE

1. The issue that causes the emergence of early worker should first of all be disputed with his immediate boss for immediate settlement.
2. If the employee is not satisfied with the decision of his immediate boss, he should notify his head of the department who will ensure the matter is settled by talking to the two parties concerned.
3. If the above failed, the issue could be discussed with the Staff Manager of the Company for settlement while those earlier connected with the issue invited.
4. If the matter remain unresolved, it can then move to the desk of the union leader for settlement.
5. If all of the above failed, the legal institution set up will now be used, so that what started as industrial grievance has not turn to collective grievance and it has not now becomes trade dispute.

MAINTENANCE:

Maintenance can be defined as the art of keeping plant or machine in good operating condition at all times with minimum cost. It means the care taken on equipment in order to carry out their various functional operations accordingly.

Preventive Maintenance is the Maintenance or Care on machine or plant carried out before failure.

OBJECTIVES OF PREVENTIVE MAINTENANCE:

- (i) To obtain maximum availability of machine or plant by avoiding break downs and by reducing the shut down periods to minimum.

- (ii) To keep the plant or machine in order condition so as to maintain the quality of the products.
- (iii) To minimise wear and tear, and to preserve the value of the plant or machine.
- (iv) To ensure the safety of the workers.
- (v) To keep the machine or plant at the maximum production efficiency.

PREVENTIVE MAINTENANCE REQUIREMENT

1. Good Supervision
2. Good Maintenance administration department.
3. Operators should be well trained
4. Good lubrication programme.
5. Proper Maintenance records.
6. Adequate stock of spare parts should be kept.
7. Correct clear and detailed instructions be given to the Maintenance crew and the operators.
8. Surroundings to be clean, ample Ventilation and illumination.
9. Manual and other data to be consulted as when required.

ADVANTAGES OF PREVENTIVE MAINTENANCE.

1. Reduction in production down time
2. Lesser overtime pay for Maintenance crew
3. Lesser number of ~~freely~~ faulty equipments
4. Lesser expenditure on repairs.
5. Lesser spare parts are needed to remain in the store ~~at~~ at all times.

RUNNING MAINTENANCE.

Running Maintenance is the Maintenance which takes place when the machine is still running.

Ex: Topping up a car battery.

BREAK DOWN MAINTENANCE.

This is carried out on a failed machine in order to restore the operational condition.
eg. engine block.

EMERGENCY MAINTENANCE.

This is the maintenance which is unforeseen and unplanned for and carried out on a machine eg. re setting of contact breaker point.

OPPORTUNITY MAINTENANCE.

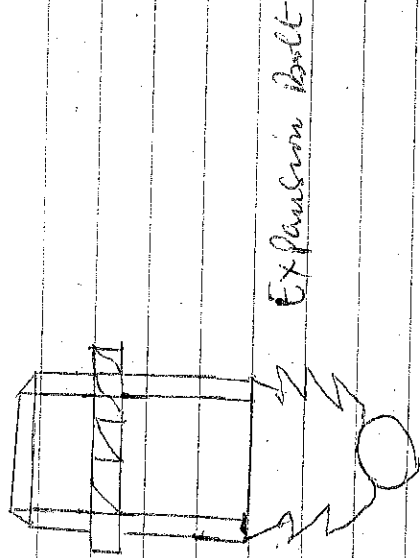
This is the maintenance carried out on a machine by chance eg. discovered worn parts after dismantling.

DESIGNANT MAINTENANCE.

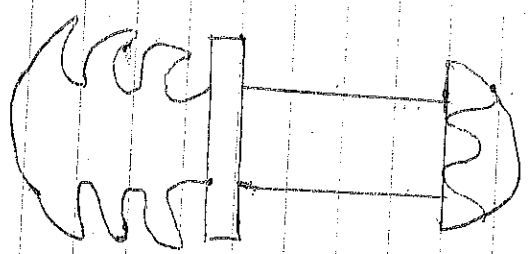
This is the maintenance carried out on a machine through modifications of the part that fails regularly eg. redesigning the components that holds the car wheel over.

TYPES OF INSTALLATION BOLTS.

a) Expansion bolt: As the nut is being tightened, the bolt stem expands in the precast hole.

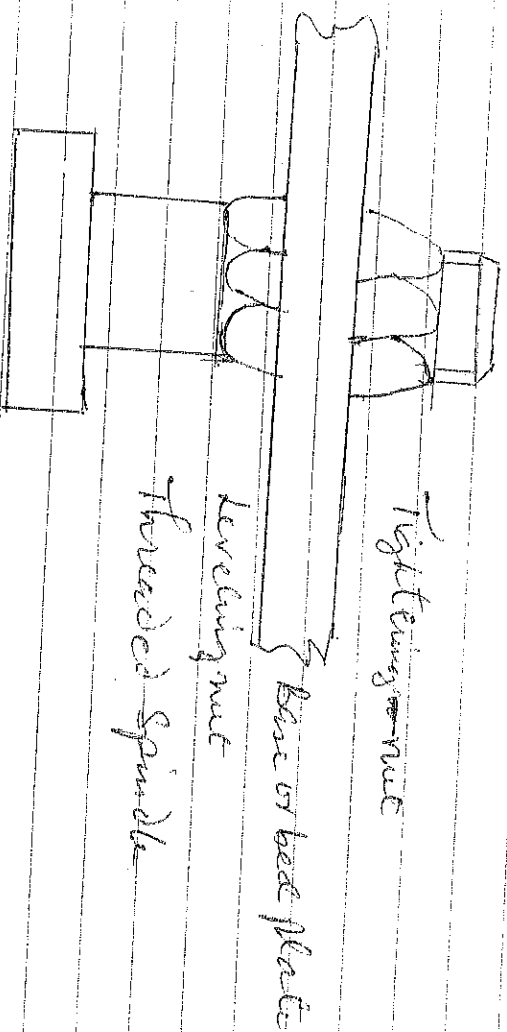


COTTER BOLT: This is kept in sleeve and it is placed inside the foundation plate. It is threaded in the same way as bolt which is screwed into these rods, whereby the pull bolt on tightening the bolt head is taken by cotter.



Cotton Bale.

VIBRATION INSULATION PADS: These are designed to Secure Machine or Plant temperature temporarily. These pads have provision for adjustment i.e. leveling the plant. The base is made of metal, but inside it, there is rubber padding. There is a threaded spindle and nuts protecting from the base. The top nut is tightening while the bottom nut is for leveling. It is necessary in order to keep the transmission vibration to Minimum.



Vibration insulator Pad.

Maintenance Monitoring

Maintenance carried out in response to a significant deterioration in unit as indicated by a change in a monitored parameter of the unit in form of signal, alarm, light colour change is called monitoring.

Methods of Condition Monitoring Technique

There are two major classifications of Condition Monitoring technique.

1. On-line Monitoring Technique
2. Off-line Monitoring Technique

On-line Monitoring Technique:- This involves the following:-

- 1) Visual and tactile inspection of accessible Monitoring points
- 2) Temperature monitoring
- 3) Lubricant monitoring
- 4) Leak detection
- 5) Vibrating monitoring
- 6) Noise monitoring
- 7) Corrosion monitoring

Off-line Monitoring Technique:- This comprises of the following:-

- 1) Visual and tactile inspection of normally inaccessible or moving parts monitoring
- 2) Cracks detection
- 3) Leak detection
- 4) Vibration testing
- 5) Corrosion monitoring

Function of work planning

The function of work planning is to get the right resources to the right place to do the right job in the right way at the right time.

Fundamentals of maintenance work planning

1. The work planner must have the authority to take the necessary decisions which affect the work load and the resources.
2. The work planner must have the right information at the right time about both the work planning and it is then necessary to follow a third rule.
3. The areas of responsibility and lines of communication between the planning levels must be clearly defined.

Characteristics of work planning

1. The deterministic load: - Is that which can both be planned and scheduled in the long term.
eg. Preventative maintenance modification projects
a portion proportion of the corrective maintenance

2. The probabilistic load: - Is that which can be scheduled only in the short term e.g. Corrective and emergency maintenance. If best, only the average level is known. The work load fluctuates in the short which can also be shown in queuing model.

Functions of maintenance dept.

1. Locate the mal-function
2. Diagnose the fault
3. Prescribe the action to correct the mal-function
4. Decide on the work priority.
5. Plan the resources necessary to undertake the work.
6. Schedule the work.
7. Issue job instructions
8. Check that the work is satisfactory.

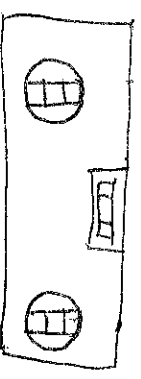
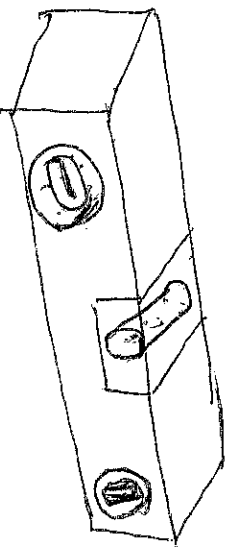
Spirit level

Definition

Types of spirit level

1. Carpenter's level (either wood, composite and aluminium materials)
 2. Mason's level
 3. Torpedo level
 4. Post level
 5. Line level
 6. Engineers precision level
 7. Electronic level
 8. Inclinometer level.
 9. Auto collimation.
- Spirit level

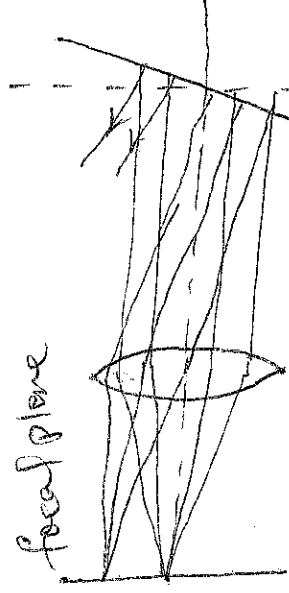
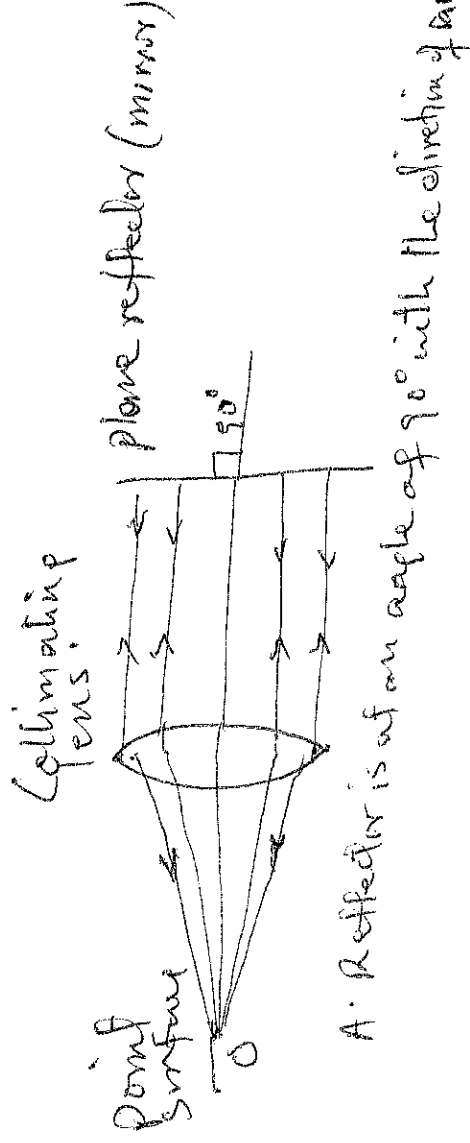
A traditional carpenter's spirit level looks like a short plank of wood and often has a wide body to ensure stability and that the surface is being measured correctly. In the middle of the spirit level is a small window where the bubble and the tube is mounted.



2. Auto collimator spirit level

An auto collimator is an optical instrument for non-contact measurement of angles, they are typically used to align components and measure deflections in optical or mechanical systems.

In the process of collimating an instrument (such as a telescope) having objective and cross hairs by directing it toward a plane mirror and subjecting the lens and cross hairs so that the latter coincide with their own reflected image.



B. Reflector is not at right angle with the direction of rays.

Leveling pad

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Leveling pad is also known as leveling foot. It is a machine device that used to level the object or machine on which it being installed. They feature an adjustable height at which the machine or object rests off the floor.

Levelers are tools used to determine whether a surface is horizontal (level) or vertical (plumb). With out this you cant build a house that works well or looks good or not to mention how taken cabinets or trim out a door.

Types of leveling instruments

1. Dumpy level
2. Y-level
3. Gualing's level
4. Tilting level
5. Costes Reversible level
6. Automatic level.

Wedges

Wedge can be defined as a piece of wood, metal etc having one thick end and tapered to a thin edge that is driven between two objects or parts of an object to secure or separate them.

Types of wedges:-

- ① Pitting wedge
- ② Gap wedge
- ③ Sand wedge and
- ④ Lat wedge.

Shim

3

A shim can be defined as a washer or thin strip of material used to align parts, make them fit or reduce wear.

Shim sometimes can be referred to as wedge or spacer.

Fill up between support, levelling or adjustment of fit.

Types of shims: - Different types of shims include

1. Arbor shims
2. Variable shims
3. Slotted shims
4. Shim rings
5. Shim stack.

All the above listed shims are used to take up excess tolerances, they each provide unique value to specific applications.

Shim materials: - Shims are made of a variety of materials

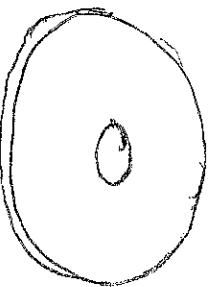
and thicknesses. MS, Plastic, aluminium, brass, stainless steel, cold-rolled or low carbon steel are good examples of shim materials. These materials meet most of the design requirements for most applications.

Standard measurements of shim sizes include;

Size A: (2x2) inch with a $\frac{5}{8}$ inch slot

Size B: (3x3) inch with a $\frac{13}{16}$ inch slot

Size C: (4x4) inch with a $1\frac{1}{4}$ slot



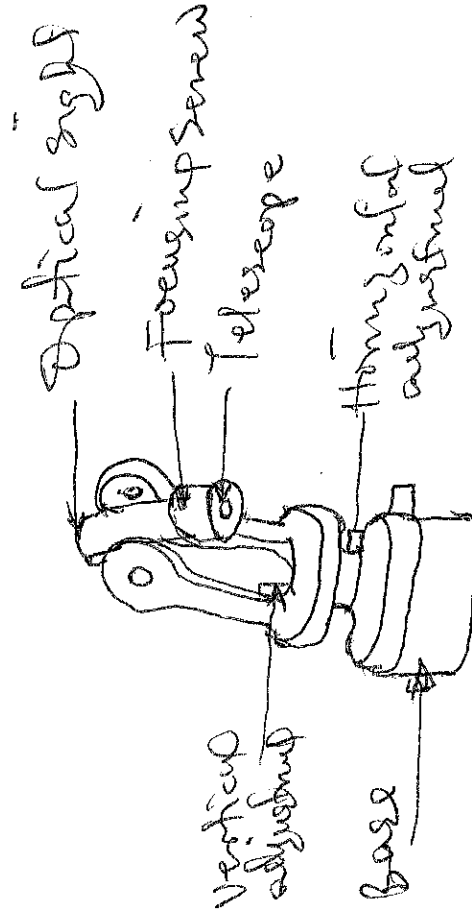
Shim.

Theodolite

A theodolite is a surveying instrument with a rotating telescope for measuring horizontal and vertical angles. In its modern form, it consists of a telescope mounted to swivel both horizontally and vertically.

There are two types of theodolite in use, viz; digital and non-digital. Non-digital type are rarely in use these days.

Digital theodolites consist of a telescope that is mounted on a base as well as an electronic readout screen that is used to display horizontal and vertical angles only. It is sometimes called tachymeters because it can calculate distance using geometric trigonometric calculations by angular measurement through a graduated sight.



THEODOLITE

Foundation Plan

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The foundation plan is a plan view drawing in section, showing the location and size of footings, piers, columns, foundation walls and supporting beams. It includes the size, type, location and spacing of all anchor bolts with the required depth of embedment, edge and end distance.

Types of foundation plan

Foundation plan are of various categories:-

1. Raft or mat footing:- This type of foundation is suitable for weak soil.
2. Combined footing:- This type is appropriate when the column is spaced closely and the soil has a low bearing capacity.
3. Deep foundations (piles):- This type of foundation is done when hard soil is very deep in the earth and far from reach.
4. Caisson or drift shaft foundations:- This occurs when the depth of hard stratum is between 25 feet and 300 feet.

Shallow foundations

This type of foundations does not exceed the safe bearing capacity of the soil. It is also called open footing, when constructing a shallow foundation you first excavate the earth, till the bottom of the footing then build the footing.

The entire footing is visible in this type of foundation. It is mostly applicable to lightweight buildings with perfect soil conditions. Shallow foundations can get damaged from freezing. Therefore, they must be protected during cold climates. Also, they should be embedded below the frost line protected from insulation.

Assignment 1

1. Prepare a foundation plan template for installation of a typical machine tool and heavy weight structure
2. Explain with detail-well labelled drawing of a standard typical Mechanical Engineering workshop.

Concrete Vibrator

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A concrete vibrator is a concrete tool that produces stronger concrete ^{by vigorously vibrating the concrete} after it is poured to eliminate the air bubbles or air pockets.

Internal vibrator machines also known as an immersion or concrete needle vibrators are commonly used on construction sites. Formwork vibrators are used on concrete columns, thin walls and precast units.

Formwork is the structure, usually temporary, used to contain poured concrete and to mould it to the required dimensions and to mould it is able to support itself. It supports until of the force contained material and the beams that directly support the face contained material.

Methods of Pumping Foundation Concrete

1. Tailgating :- Tailgating is the method of pumping concrete straight down the chute from the concrete truck.
2. Concrete buggies :- This is also called Georgia buggies and power buggies.
3. Pumping :- There are two main types of concrete pumps.
 4. Boom pumps
 5. line pumps
 6. excavator bucket

Steps for pumping foundation concrete

1. Excavation
2. Footings (Formwork and concrete pour)
3. Walls (Formwork and concrete pour)
4. Water proofing (Concreditions, Bituminous coating, Crystalline etc)
5. Drainage pipe/weeping tile
6. Back fill
7. Paving/finishing concrete floor.

Opportunity maintenance: - Is the maintenance carried out on equipment by chance in the process of carrying out maintenance of a machine or plant.

ex. Change of oil seals and gaskets during engine overhauling.

Running Maintenance: This is the type of maintenance usually carried out on machine or plants while it is working without necessary stopping the plant completely from operation.

MAINTENANCE PLAN

- Maintenance plan generally involves making provisions such as
- (i) Adequate and well trained maintenance personnel
 - (ii) Make available money necessary for maintenance, then, where and when to get the money will have to be stated.
 - (iii) All materials and replacement parts must be made available including toolings. Replacement parts could be made in house or purchased from market. These have to be specified and machines / equipment required must be specified and
 - (iv) Adequate space must be put into consideration.
 - v. Above all, do not keep un-necessary inventory. Apply part analysis in your maintenance plan.

MAINTENANCE PROCEDURES

Maintenance procedures should be daily and periodic. Daily maintenance is done by operators themselves which include daily cleaning, oiling, greasing etc. Before the operator starts work either normal or shift, daily maintenance is essential. Periodic maintenance is conducted by the maintenance crew based on past experiences and operating instructions found in the manuals. During these periods both inspections and lubrication schedules should be strictly followed and any defect to be rectified immediately.

All actions necessary to return or restore an equipment to an acceptable condition is ~~called~~ termed as maintenance. In another word, it means care. i.e. the care that is taken on equipment in order to carry out their various functional operations accordingly. Maintenance can be classified as follows;

- (i) Preventive maintenance
- (ii) Schedule maintenance
- (iii) Break down maintenance
- (iv) Emergency maintenance
- (v) Opportunity Maintenance
- (vi) Running Maintenance.

Preventive maintenance: - This is a type of maintenance usually carried out on plant in order to prevent failure.

Schedule maintenance: - Is the maintenance or care carried out machine or plant with fore knowledge of control and records which may be daily, weekly, monthly, annually or bi-annually.

Breakdown maintenance: - Is a maintenance carried on equipment when it has failed completely in order to restore it back to its normal operational condition. eg replacement of a broken crankshaft.

Emergency maintenance: - Is the care carried out on machine with a fore knowledge of the breakdown. This maintenance is usually carried out immediately the machine fails in order to restore it to normal operational condition. eg. Setting of a contact breaker point.

The ponds department can be termed to be the backbone of any Service Station or repair workshop as ~~all~~ services demands the use of ponds during maintenance work on vehicles. It is the duty of the ponds department to Service the workshop maintenance section with all the necessary required ponds that may be needed to meet up their customers requirements during maintenance work.

A reasonable size of Steele has to be monitored by the ponds department. The running out of Steele will mean a loss to the organization Capital tightening down which also means a loss to the organization. Therefore a Steele even pond has to be reached for which ponds can be maintained in Steele.

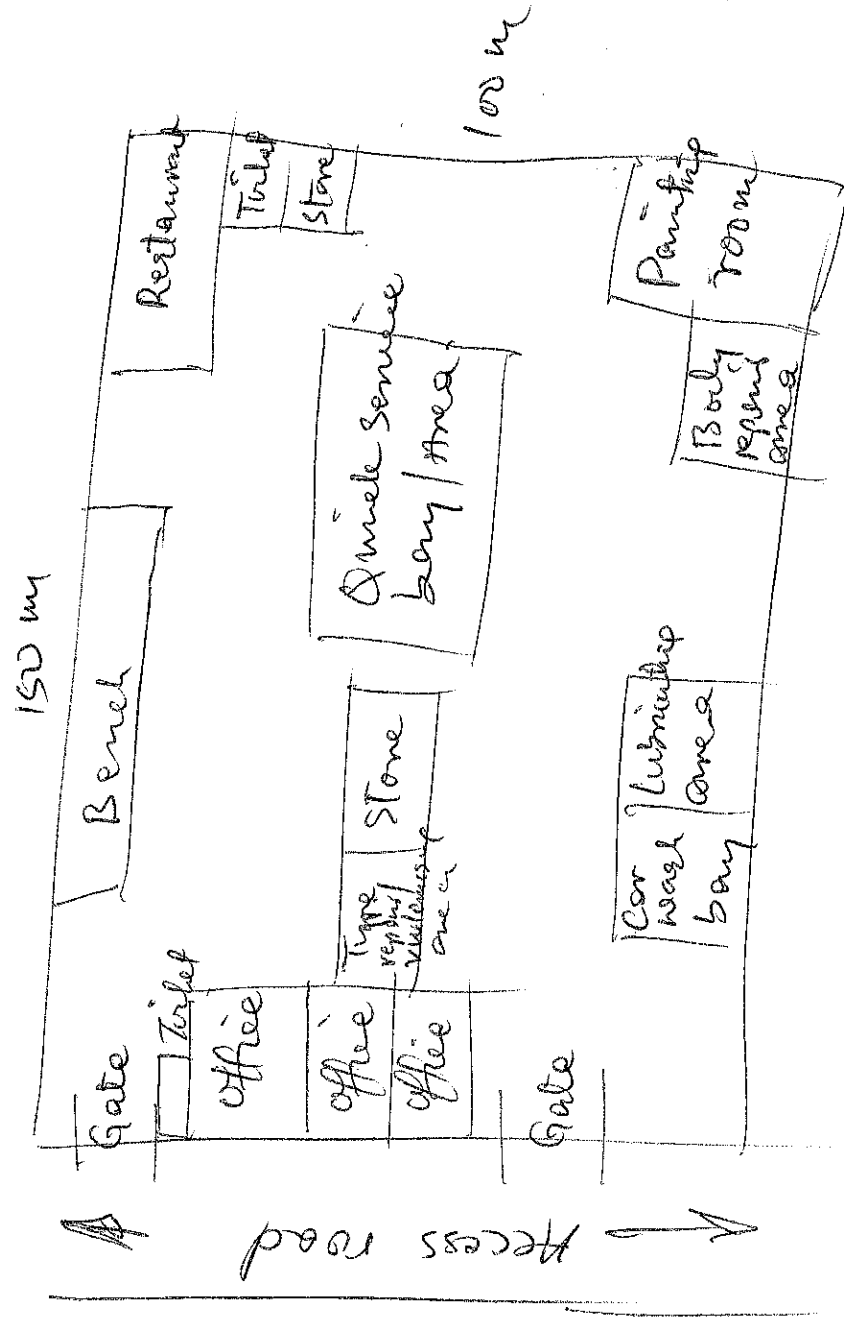
Workshop layout

The layout of any workshop will depend entirely upon the work anticipated or work already being completed in the case of an existing Service Station.

Small workshops usually have a bench at the closed end of the workshop with one pit or underneath a car. In this modern age, small workshops have a very limited volume as modern vehicles demand sophisticated equipment with rapidly diagnose faults and means of quality control after repair.

To ensure a profitable workshop, floor space must be used to the optimum so that maximum profit can be achieved.

In a developed country, workshop floor space is very important in that it is being used sometimes for rental purposes. Cars are being charged in the space per day. This sometimes entices down vehicles report work on broken down the repairs which awaits repairs so that additional cars shall not experience paying



Workshop layout.

While the service manager may deal with workshop bookings, the reception engineer will need to know when a job can be completed in the workshop. To load a workshop with too much work means a customer is left without his car for too long period whilst too little work means mechanics stand idle. Thus a daily diary should be kept and full details of the customers name and address and the jobs required.

If a repair job is needed, the customer should be checked upon arrival and the job assessed. e.g. normal servicing jobs from minor oil changes and minor schedules when keeping the diary, the reception engineer should keep in very close liaison with the foreman to see how jobs are progressing and whether full staff is available or if any are off-ill, going on courses before accepting a job should be in mind the following:

- a) Number of mechanics in the workshop e.g. 10 mechanics for 8 hours working day 80 hrs per day working time.
- b) Type of jobs coming in (repair/service) and length of time required manufacturers stand and line schedule for repair work should be consulted.

(C, Must convenient time to accept jobs unless customers can have a car all day. Rental kept for flow spare should always be kept in mind.

d, If points are required of time required approximately length whether they can be ordered before the job comes in. Liaison with time manager is essential for this part of work.

e, Upon arrival of the job, a job card should be issued made out in duplicate, one to the office, one for the foreman to pass into the machine unless a three or five point job set is used where one copy is the job card.

SERVICE

Service may be defined from repair based in the following.

Service may be defined as the day to day maintenance that one usually given to a machine or equipment so as to keep its condition and overall performance effectively. Service are always scheduled within interval of time. eg. changing of engine oil or filter.

while repair refers to, change of parts or possibly reconditioning the engine like an effective one. Repair has its time schedule like service whenever it fails, parts and maintenance work has to be administered to restore it back to its normal operating condition eg. change of rings, piston, crankshaft etc.

RESPONSIBILITIES OF RECEIVING EXTENSION

1. He meets the customers and assess the fault with the vehicle by using equipment and or by driving tests, obtain customers approval on the repairs to be made and place order for parts if need be.
2. Having obtain approval from the customer, repair and passed out an estimate for the service in order paper.
3. He revised the job card and passes them to the required destinations.

4. He keeps a record of workshop loadings and checked on daily basis with the workshop foreman in case a mechanic or his staff to show up for work - eg illness.

5. He prepares the invoice of bill and forward it to the account dept. of the garage. He hand over the car to the customer after service or repair of the final end.

ESTIMATE FOR ACCIDENT VEHICLE REPAIR

BEN AUTOMOTIVE REPAIRS WORKSHOP

Branch office
No 14 Bonoma layout
Mubbi, north
Adama area state
Tel:

Head office
No. 4. Xx close
Idumaba Lagos
Tel:

ESTIMATE FOR THE REPAIR OF HEAD-ON COLLISION CAR

Veh. no.		Veh. Type		Mileage	
Chassis no.				Policy no.	
S/no	Description of item	Qty	Price each	Amount	
1.	Bumper	1 pc	40,000	40,000	
2	Windscreens	1 pc	15,000	15,000	
3	Front	2 pcs	16,000	32,000	
4	Headlamps	1 set	30,000	30,000	
5	Radiator	1 pc	12,000	12,000	
6	Radiator grill	1 set	5,000	5,000	
7	body work	Complete	40,000	40,000	
8	Painting	As required	30,000	30,000	
9	VAT	5%	30,000	30,000	
Total			5%	30,000	5%
Customer's Sign		Engineers Sign		Manager's Sign	